# Chapter 2. Outliers and Robust Prediction

Predictive Modeling, Section 2 (MCD, LOF, Isolation Forest, leverage, Cook's distance, Huber, RANSAC, Theil–Sen)

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# 1 An Outlier Is Not the Same as a Data Error

| What the observation is | Example | First response |
|---|---|---|
| input / measurement error | wrong unit, mis-recorded target | correct from the raw source; remove only if uncorrectable |
| a genuine rare case | a very large transaction, a rare high-risk patient | keep it, and check that group's prediction error separately |
| a case from a new environment | a policy or market condition absent in training | check distribution shift and the need to retrain |

An outlier is a *statement relative to a reference*, and the right action depends on why it is unusual.

## 1.1 Three kinds of outliers

| Kind | Meaning | Simple diagnostic |
|---|---|---|
| univariate | one variable is far from most observations | histogram, boxplot, IQR rule |
| multivariate | each variable is ordinary but the *combination* is rare | scatterplot, multivariate distance, density |
| conditional | given $x_i$, the target $y_i$ differs greatly from what is expected | residual $e_i = y_i - \hat{y}_i$ |

The IQR rule flags $x < Q_1 - 1.5\,\mathrm{IQR}$ or $x > Q_3 + 1.5\,\mathrm{IQR}$, $\mathrm{IQR} = Q_3 - Q_1$. Univariate rules alone miss outliers that appear only in combinations of variables.

| Method | What it checks | Observations it looks for |
|---|---|---|
| MCD | global distance in $X$ | multivariate outliers far from the central distribution |
| LOF | local density in $X$ | points less dense than their neighbors |
| Isolation Forest | isolation path in $X$ | points isolated by few random splits |
| studentised residual | $y$ given $X$ | cases far from the regression prediction |
| leverage, Cook's distance | influence on the regression fit | cases that change coefficients and predictions a lot |

**"$X$ is unusual" and "$y$ is unusual given $X$" are different problems.**

# 2 MCD: Robust Mahalanobis Distance

## 2.1 Classical Mahalanobis distance and masking

With the sample mean $\bar{x}$ and sample covariance $S$,

$$D_i^2 = (x_i - \bar{x})^\top S^{-1} (x_i - \bar{x}).$$

**Theorem: null distribution of the squared distance**

If $X \sim N_p(\mu, \Sigma)$ then $(X-\mu)^\top \Sigma^{-1}(X-\mu) \sim \chi^2_p$. Hence, approximately, a point with $D_i^2 > \chi^2_{p,1-\alpha}$ is an outlier candidate.

**Proof.** $Z = \Sigma^{-1/2}(X-\mu) \sim N_p(0, I_p)$, and $(X-\mu)^\top\Sigma^{-1}(X-\mu) = Z^\top Z = \sum_{j=1}^p Z_j^2$, a sum of $p$ independent squared standard normals. ■

**Masking.** Outliers pull $\bar{x}$ toward themselves and inflate $S$, so their own distances shrink and they may no longer be flagged. The remedy is to estimate location and scatter from the bulk of the data only.

### 2.2 Minimum Covariance Determinant

**Definition: MCD**

For a fixed $h$ with $n/2 \le h < n$ (typically $h \approx 0.75n$), find the subset $H$ of $h$ observations whose sample covariance has the smallest determinant:

$$H^\star = \arg\min_{|H|=h} \det S_H, \qquad \hat{\mu}_{\mathrm{MCD}} = \bar{x}_{H^\star}, \quad \widehat{\Sigma}_{\mathrm{MCD}} = c_{h,n}\, S_{H^\star},$$

($c_{h,n}$ is a consistency factor), and use the robust distance

$$D_i^2 = (x_i - \hat{\mu}_{\mathrm{MCD}})^\top \widehat{\Sigma}_{\mathrm{MCD}}^{-1} (x_i - \hat{\mu}_{\mathrm{MCD}}), \qquad \text{flag if } D_i^2 > \chi^2_{p,1-\alpha}.$$

$\det S_H$ is proportional to the squared volume of the data ellipsoid of $H$, so MCD looks for the *tightest* $h$ points; outliers enlarge the volume and are excluded. Up to $n-h$ outliers cannot drive the estimate arbitrarily far (breakdown point $\approx (n-h)/n$). In practice the FAST-MCD algorithm iterates "C-steps": given $H$, compute distances and replace $H$ by the $h$ points with smallest distances; each step does not increase $\det S_H$.

**Remark: when MCD fits**

MCD suits continuous data with $p < n$ and a single elliptical bulk. With multimodal or nonlinear clouds, local methods (LOF) or Isolation Forest are more appropriate.

## 3 LOF: Lower Density Than the Neighbors

**Definition: local outlier factor**

Let $N_k(i)$ be the $k$ nearest neighbors of $i$, $d_k(j)$ the distance from $j$ to its $k$th neighbor, and the reachability distance $\mathrm{rd}_k(i,j) = \max\{d_k(j), d(i,j)\}$. The local reachability density and LOF are

$$\mathrm{lrd}_k(i) = \Big(\frac{1}{|N_k(i)|} \sum_{j \in N_k(i)} \mathrm{rd}_k(i,j)\Big)^{-1}, \qquad \mathrm{LOF}_k(i) = \frac{1}{|N_k(i)|} \sum_{j \in N_k(i)} \frac{\mathrm{lrd}_k(j)}{\mathrm{lrd}_k(i)}.$$

$\mathrm{lrd}_k(i)$ grows as the distances to the neighbors shrink. LOF $\approx 1$: density similar to the neighbors; LOF $> 1$: lower density than the neighbors; very large LOF: a local outlier separated from its surrounding group. Because the comparison is *relative to the neighborhood*, a point in a sparse but genuine cluster is not penalized, unlike a global distance rule. The reachability distance smooths $d(i,j)$ for very close pairs, which stabilizes the ratio.

# 4 Isolation Forest: Quickly Isolated Points

**Algorithm: Isolation Forest**

1. On a subsample, choose a feature uniformly at random.
2. Choose a split point uniformly between the feature's minimum and maximum.
3. Split recursively until each observation is alone (or a depth limit).
4. Grow many such trees and average the path lengths.

Anomaly score:

$$s(x) = 2^{-\mathbb{E}[h(x)]/c(n)}, \qquad c(n) = 2H(n-1) - \frac{2(n-1)}{n}, \quad H(m) \approx \ln m + 0.5772,$$

where $h(x)$ is the path length until $x$ is isolated and $c(n)$ is the average path length of an unsuccessful search in a binary search tree of $n$ points (normalization by sample size).

Short paths give large scores: $s \to 1$ means anomalous, $s \approx 0.5$ for all points means no clear anomaly, $s \ll 0.5$ means normal. No distributional shape is assumed, and it scales to large data. The parameter `contamination` only sets the share of points to be labeled anomalous. Isolation Forest gives a score, but it does not tell a data error from a valid rare case.

# 5 Regression: Residual, Leverage and Influence

## 5.1 Three different notions

Suppose most data follow $y \approx 2x$.

| Observation | Feature $x$ | Residual | Interpretation |
|---|---|---|---|
| $(3, 20)$ | ordinary range | large | outlier in the target |
| $(10, 20)$ | far from the rest | small | high-leverage point that follows the trend |
| $(10, 0)$ | far from the rest | large | influential point likely to change the line |

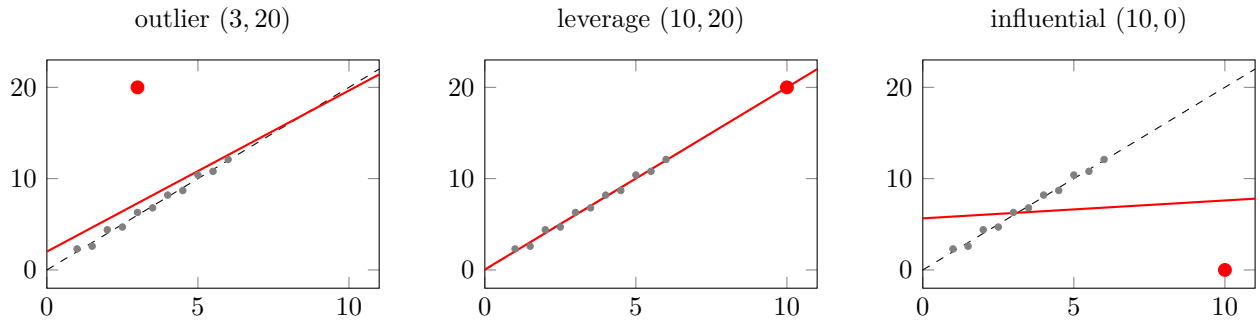



**Figure 1:** Dashed: OLS line on the gray points ($y \approx 2x$); red: OLS line after adding the red point (exact refits). A target outlier in the middle of the $x$-range mostly shifts the line upward; a high-leverage point on the trend changes nothing; a high-leverage point off the trend drags the slope — large residual *and* large leverage, hence large Cook's distance.

The residual is the gap between truth and prediction; leverage measures how unusual $x_i$ is in feature space; an influential point is one whose removal changes the fit a lot.

## 5.2 Hat matrix and leverage

With $X \in \mathbb{R}^{n\times(p+1)}$ (intercept included) and $H = X(X^\top X)^{-1}X^\top$, $\hat{y} = Hy$ and $e = (I-H)y$. The leverage is $h_{ii} = x_i^\top (X^\top X)^{-1} x_i$.

**Theorem: properties of leverage and residuals**

1. $0 \le h_{ii} \le 1$ and $\sum_i h_{ii} = p+1$, so the average leverage is $(p+1)/n$.
2. $\hat{y}_i = \sum_j h_{ij} y_j$, so $h_{ii} = \partial \hat{y}_i / \partial y_i$.
3. Under $\mathrm{Var}(\varepsilon) = \sigma^2 I$, $\mathrm{Var}(e_i) = \sigma^2(1-h_{ii})$.

**Proof.** $H$ is symmetric idempotent, so $h_{ii} = (H^2)_{ii} = \sum_j h_{ij}^2 \ge h_{ii}^2$, giving $0 \le h_{ii} \le 1$; $\sum_i h_{ii} = \mathrm{tr}H = \mathrm{tr}\{(X^\top X)^{-1}X^\top X\} = p+1$. Item 2 is $\hat{y} = Hy$. For 3, $e = (I-H)\varepsilon$ because $(I-H)X = 0$, so $\mathrm{Var}(e) = \sigma^2(I-H)(I-H)^\top = \sigma^2(I-H)$. ■

A high-leverage point pulls the fitted line toward itself, so its *raw* residual tends to be small. Item 3 is why residuals must be standardized.

## 5.3 Studentised residual and Cook's distance

| Diagnostic | Computation | Cases it reveals |
|---|---|---|
| studentised residual | $r_i = \dfrac{e_i}{\hat{\sigma}\sqrt{1-h_{ii}}}$ | $y_i$ far from what $x_i$ predicts |
| leverage | $h_{ii}$, diagonal of $H$ | far from the other observations in feature space |
| Cook's distance | $D_i = \dfrac{r_i^2}{p+1}\dfrac{h_{ii}}{1-h_{ii}}$ | removing it changes coefficients and predictions much |

Common screening rules are $|r_i| > 2$, $h_{ii} > 2(p+1)/n$, $D_i > 4/n$. Exceeding a rule makes a case *an object of inspection*, not an automatic deletion.

**Theorem: leave-one-out update and Cook's formula**

Let $\hat{\beta}_{(i)}$ be the OLS estimate without case $i$. Then

$$\hat{\beta} - \hat{\beta}_{(i)} = \frac{(X^\top X)^{-1} x_i e_i}{1-h_{ii}},$$

and Cook's distance, defined as $D_i = \dfrac{\|X\hat{\beta} - X\hat{\beta}_{(i)}\|^2}{(p+1)\hat{\sigma}^2}$, equals $\dfrac{r_i^2}{p+1}\dfrac{h_{ii}}{1-h_{ii}}$.

**Proof.** Let $A = X^\top X$. Removing row $i$ gives $A_{(i)} = A - x_i x_i^\top$ and $X_{(i)}^\top y_{(i)} = X^\top y - x_i y_i$. Sherman–Morrison: $A_{(i)}^{-1} = A^{-1} + \frac{A^{-1}x_i x_i^\top A^{-1}}{1-h_{ii}}$. Then

$$\hat{\beta}_{(i)} = \Big(A^{-1} + \tfrac{A^{-1}x_i x_i^\top A^{-1}}{1-h_{ii}}\Big)(X^\top y - x_i y_i) = \hat{\beta} - A^{-1}x_i y_i + \frac{A^{-1}x_i(\hat{y}_i - h_{ii}y_i)}{1-h_{ii}} = \hat{\beta} - \frac{A^{-1}x_i(y_i - \hat{y}_i)}{1-h_{ii}}.$$

Hence $\|X(\hat{\beta} - \hat{\beta}_{(i)})\|^2 = \frac{e_i^2}{(1-h_{ii})^2} x_i^\top A^{-1} X^\top X A^{-1} x_i = \frac{e_i^2 h_{ii}}{(1-h_{ii})^2}$. Divide by $(p+1)\hat{\sigma}^2$ and substitute $e_i^2 = r_i^2\hat{\sigma}^2(1-h_{ii})$. ■

So Cook's distance is large only when *both* the standardized residual and the leverage factor $h_{ii}/(1-h_{ii})$ are sizeable, which is exactly the case $(10, 0)$ above.

**Remark: leave-one-out residuals for free**

The same identity gives $y_i - x_i^\top \hat{\beta}_{(i)} = e_i/(1-h_{ii})$, so the leave-one-out CV error of OLS is $\frac{1}{n}\sum_i \{e_i/(1-h_{ii})\}^2$ without refitting.

# 6 What to Do With Detected Cases

| Finding | Action | Caveat |
|---|---|---|
| input / measurement error | correct or remove in raw data | remove only if it cannot be corrected and cannot be used |
| heavily skewed variable | log or Box–Cox transform | keeps the order, compresses gaps among large values |
| extremes must be capped | winsorisation | cap both tails at quantiles *estimated on training data* |
| valid rare case | keep | use robust learning and segment-wise evaluation |

Never delete or alter observations on the basis of a detection score alone.

# 7 Robust Learning

| Method | Core principle | Suitable when |
|---|---|---|
| Huber regression | caps the growth rate of loss for large residuals | a few large target outliers |
| RANSAC | fits many small subsamples; keeps the model supported by most points | clear inlier vs. contaminated groups |
| Theil–Sen | median of pairwise slopes | linear regression with few features |

## 7.1 Huber loss

With residual $r = y - \hat{y}$:

| Loss | Definition | Large residual | Target of prediction |
|---|---|---|---|
| squared | $r^2$ | grows quadratically | conditional mean |
| absolute | $\lvert r\rvert$ | grows linearly | conditional median |
| Huber | quadratic for small, linear for large $\lvert r\rvert$ | linear beyond $\delta$ | compromise between center fit and tail robustness |

$$\rho_\delta(r) = \begin{cases} \frac{1}{2}r^2, & |r| \le \delta, \\ \delta(|r| - \frac{1}{2}\delta), & |r| > \delta, \end{cases} \qquad \psi_\delta(r) = \rho_\delta'(r) = \begin{cases} r, & |r| \le \delta, \\ \delta\,\mathrm{sign}(r), & |r| > \delta. \end{cases}$$

**Derivation: why Huber limits outliers — IRLS view**

The estimating equation is $\sum_i \psi_\delta(y_i - x_i^\top\beta)x_i = 0$. Writing $\psi_\delta(r) = w(r)\,r$ with $w(r) = \min\{1, \delta/|r|\}$ turns it into weighted least squares $\sum_i w_i(y_i - x_i^\top\beta)x_i = 0$: iterate $\beta \leftarrow (X^\top W X)^{-1} X^\top W y$ with $W = \mathrm{diag}\{w(r_i)\}$. A case with residual $10\delta$ gets weight 0.1; its pull on the fit (the influence $\psi$) is bounded by $\delta$, whereas under squared loss it grows without bound. The function $\rho_\delta$ is convex and continuously differentiable, so the minimizer

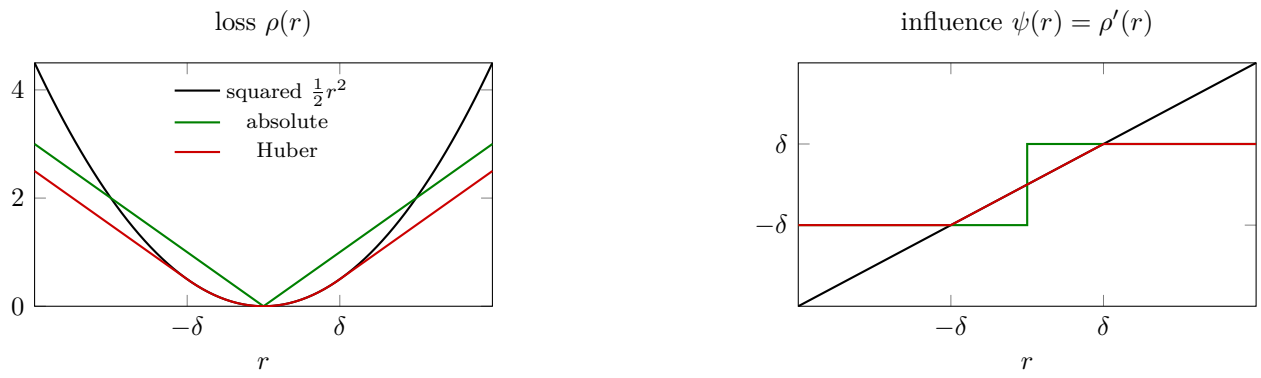



**Figure 2:** Squared, absolute and Huber losses (left) and their derivatives (right). The influence of a residual on the estimating equation grows without bound under squared loss but is capped at $\pm\delta$ under Huber loss.

is well defined. A smaller $\delta$ limits large residuals more strongly; as $\delta \to \infty$ Huber becomes OLS, as $\delta \to 0$ it behaves like LAD (rescaled).

Since $\delta$ is in units of $y$, it is usually set as $\delta = 1.345\hat{s}$ with a robust scale $\hat{s}$ (e.g. MAD/0.6745); 1.345 gives 95% efficiency under Gaussian errors.

## 7.2 RANSAC

**Algorithm: RANSAC**

1. Fit the regression on a small random subsample (minimal size $s$).
2. Classify observations with residual below a threshold as inliers.
3. Repeat many times.
4. Refit the model on the largest inlier set.

If a fraction $w$ of points are inliers, a subsample of size $s$ is all-inlier with probability $w^s$, so to succeed at least once with probability $q$ one needs $N \geq \log(1-q)/\log(1-w^s)$ trials. The residual threshold and the number of trials affect the result.

## 7.3 Theil–Sen

In simple regression compute all pairwise slopes:

$$\hat{\beta}_1 = \underset{i<j}{\text{median}} \frac{y_j - y_i}{x_j - x_i}, \qquad \hat{\beta}_0 = \underset{i}{\text{median}}(y_i - \hat{\beta}_1 x_i).$$

A few extreme slopes cannot dominate the estimate; the breakdown point is about $1 - 1/\sqrt{2} \approx 29\%$. RANSAC separates inliers and outliers sharply; Theil–Sen robustifies continuously through the median of slopes.

## 7.4 Quantile regression

The pinball loss of Ch. 1 gives the conditional $\tau$-quantile; $\tau = \frac{1}{2}$ is LAD (least absolute deviations). It is robust to outlying $y$ (not to high-leverage $x$) and describes the conditional distribution beyond the center.

## 8 Robust Prediction Is Also Chosen by CV

**Algorithm: protocol**

1. Split off the final test set first.
2. Inside each CV training fold, fit scaling and outlier-detection steps.
3. Set the detection threshold, LOF's $k$, Huber's $\delta$, RANSAC's residual threshold from training folds only.
4. Compare OLS and robust regressions on the same validation folds.
5. Check errors of outlier candidates and of rare groups along with the overall error.

| Result to check | Purpose |
|---|---|
| MAE and RMSE | evaluate typical errors and rare large errors together |
| sensitivity to the detection rule | does model selection change with the threshold? |
| error of rare groups | did robustification ignore important cases? |

## Summary

- Outliers are relative to a reference; errors, valid rare cases and new environments call for different actions.
- In $X$: MCD (global, elliptical), LOF (local density), Isolation Forest (isolation depth).
- In regression: residual ($y$ given $x$), leverage ($x$ unusual), Cook's distance (both together).
- Robust losses (Huber, LAD, quantile) and robust fitting (RANSAC, Theil–Sen) limit the dominance of a few points; their tuning constants are chosen inside CV.